CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Pharmacy (B. Pharm)

Semester – II University Backlog Theory Examination October-2015

PH108 Pharmaceutical Chemistry-II (Physical)

Date: 16/10/2015, Friday Time:10:00 a.m To 01:00 p.m Maximum Marks: 80

Instructions:

1. **Figures to right indicate marks.**
2. **Section wise separate answer book is to be used.**
3. **Draw neat sketches wherever necessary.**

SECTION- I

Q 1		**Attempt any TWO of the following:**	
A		Enlist the Physical properties of Liquid states. Explain any one physical property with its measurement methods.	**10**
B		**Answer the Following:**	
	1.	Short note on Heterogeneous Catalysis reaction.	**05**
	2.	Acid-base catalysis with suitable examples.	**05**
C		Explain Application of Radiopharmaceutical Products with Examples.	**10**
Q 2		**Attempt any TWO of the following:**	
A		Give detail construction, working and measurement of Geiger Muller counter method of analysis for radio products.	**10**
B		**Write a Short Note on:**	
	1.	Jablonski diagram with its importance	**05**
	2.	Quantum efficiency with examples.	**05**
C		**Answer the Following:**	
	1.	Radiopharmaceutical Products.	**05**
	2.	Characteristics of Enzyme Catalysis.	**05**

SECTION- II

Q 3		**Attempt any TWO of the following:**	
A		What is surface tension? Write about various methods for the estimation of surface tension.	**10**
B		What is mean by Phase rule? Write a brief note on One component three phase system in detail.	**10**
C		**Write a Short Note on:**	
	1.	Joule-Thomson effect.	**05**
	2.	Debye-Huckel theory	**05**
Q 4		**Attempt any TWO of the following:**	
A		Write a note on Carnot cycle and derive equation of thermodynamic efficiency.	**10**
B		**Answer the Following:**	
	1.	Justify: Calcium is an important electrolyte of our body.	**05**
	2.	Importance of Magnesium as an electrolyte for body.	**05**

C **Answer the Following:**

1. Differentiate Ideal Solution and Real solution **05**
2. Explain following terms: **(Any FIVE)** **05**

[1] Parachor, [2] Boiling Point, [3] Conductance,[4] Interstitial Fluid,
[5] Viscosity [6] Isothermal Process [7] Optical Rotation